Predicting U.S. Domestic Flight Delays: MLP and Bayesian Logistic Regression

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Introduction/Abstract

Flight delays cost the U.S. aviation industry billions annually and affect millions of passengers. Cascading disruptions mean a single delayed flight can ripple through an entire day's schedule.

Research Question: Can we predict whether a U.S. domestic flight will depart 15 or more minutes behind schedule, and what factors drive that risk?

We compare two approaches:

- MLP Neural Network (Python/Keras)
- Bayesian Logistic Regression via manual Metropolis-Hastings (R)

Data: BTS On-Time Performance, Jan–Dec 2025
Source: `transtats.bts.gov`

- 6,879,484 flights | 14 carriers | 352 airports
- 21.7% delayed | 78.3% on time
- `github.com/anc002/flight-delay-prediction`

Motivation and Data

Prior work shows that temporal variables and network-level congestion are strong predictors of departure delays (Rebollo & Balakrishnan, 2014; Hatıpoğlu & Tosun, 2024). Deep learning further improves accuracy when combined with weather data (Qu et al., 2020). This project establishes a schedule-only performance ceiling using features available before departure.

Key EDA Findings:

- Delay rates peak in June–July (∼27%) and drop in September (∼16%)
- Evening departures (hours 19–22) exceed 30% delay rate
- Sunday and Monday are the highest-delay days (∼24%)
- Frontier, Allegiant, and Spirit show the highest carrier delay rates
- Top delay airports include ASE, SFB, DFW, DAL, and PBI

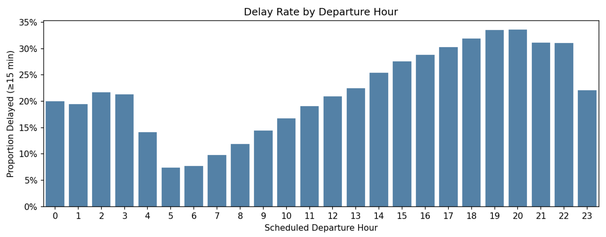

Fig. 1: Delay rate by departure hour.

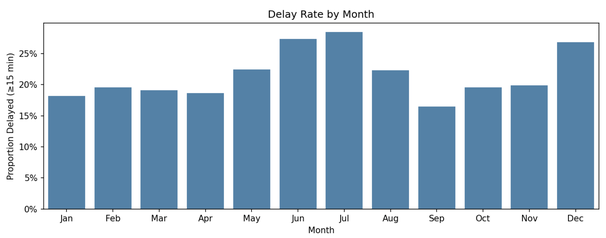

Fig. 2: Delay rate by month.

Methodology

Cleaning: The raw dataset contained 7 million rows. Cancelled and diverted flights were removed as they cannot be delayed in the traditional sense. The scheduled departure time was converted to a departure hour feature. After cleaning, 6,879,484 usable records remained with no missing values.

MLP (Python/Keras)
Architecture: Input (133 features) → Dense(64, ReLU) + Dropout(0.3) → Dense(32, ReLU) + Dropout(0.3) → Dense(1, Sigmoid). Trained on 480k flights from a stratified 600k subsample (80/20 split). Adam optimizer, binary cross-entropy loss. Class weights (0.64 on-time / 2.31 delayed) applied to correct for 78/22 class imbalance. Top-50 airports one-hot encoded; numeric features standardized on training set only.

Bayesian Logistic Regression (R)
Bernoulli likelihood with sigmoid link and $\mathcal{N}(0, 10)$ priors on all coefficients, weakly informative and placing no strong directional belief on any feature. Fit via manual Metropolis-Hastings: 100,000 iterations, 18% burn-in, thinned by 50, proposal $\sigma = 0.01$ yielding approximately 27% acceptance rate. Trained on a stratified 50k subsample due to MCMC scaling constraints. Top-20 airports encoded.

Results

The MLP achieved a **ROC-AUC of 0.7154** and successfully identified 63% of actual delays. Precision on the delayed class was 35%, reflecting a deliberate recall-bias from class weighting. Missing a delay is more costly than a false alarm in practice, so the model was tuned to prioritize recall. Training loss converged steadily across 50 epochs with no overfitting observed.

The Bayesian model achieved a **ROC-AUC of 0.6694**. The DEP_HOUR 95% credible interval [0.49, 0.54] lies entirely above zero, confirming departure hour as a reliable positive predictor. Distance and carrier coefficients show wider intervals overlapping zero, suggesting weaker individual signal. The Bayesian model produces full probability distributions over delay risk, supporting uncertainty-aware decisions.

Fig. 3: MLP training vs. validation loss.

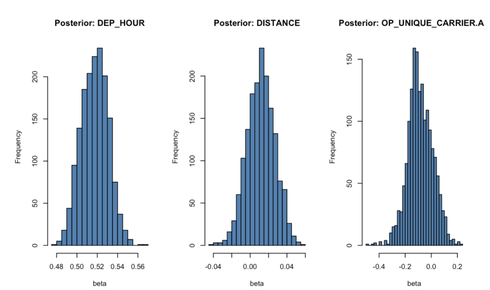

Fig. 4: Posterior distribution of DEP_HOUR, DISTANCE and Alaska Airlines coefficients.

Discussion and Future Work

This project demonstrates the complementary strengths of neural network and Bayesian approaches to delay prediction. The MLP offers higher raw discriminative performance, capturing non-linear interactions between departure hour, carrier, and airport that a linear model cannot. The Bayesian model, while trained on a smaller subsample due to MCMC scaling constraints, provides something the MLP cannot: a full posterior distribution over each coefficient, allowing us to quantify how confident we are that a given feature actually drives delays.

The gap in ROC-AUC between the MLP (0.7154) and Bayesian model (0.6694) is expected. The Bayesian model was trained on 50k observations versus 480k for the MLP, and uses a linear decision boundary that cannot capture non-linear feature interactions. The value of the Bayesian model lies not in raw discrimination but in uncertainty quantification. A model that tells you how confident it is in each prediction is more useful in practice than a black-box model with marginally higher AUC.

Both models performing in the 0.67–0.71 AUC range without weather data is itself a finding consistent with the literature. Qu et al. (2020) showed that adding meteorological data significantly boosts accuracy. Our results confirm that schedule-based features alone approach but do not exceed that ceiling.

Model Comparison:

	MLP	Bayesian
ROC-AUC	0.7154	0.6694
Recall (delayed)	63%	–
Precision (delayed)	35%	–
Training data	480k	50k
Uncertainty estimates	No	Yes

Limitations:

- No weather data available at time of prediction
- Prior-flight delay propagation via tail number not included
- MCMC subsampling to 50k limits exposure to rare airport and carrier combinations

Future Work:

- Integrate NOAA weather observations at departure airports
- Encode prior-flight delay as a feature via tail number joins
- Apply variational inference for scalable Bayesian modeling on the full 6.8M dataset
- Deploy a real-time delay probability tool using the trained MLP

References

Rebollo & Balakrishnan (2014). Transportation Research Part C, 44, 231–241.

Hatıpoğlu & Tosun (2024). Applied Sciences, 14(13), 5472.

Qu et al. (2020). Neural Processing Letters, 52, 1–19.